

## Experiment 04

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### Objective:

To introduce the components used in a Radar Detection System and understand their basic functionality with Arduino Uno. The experiment will focus on setting up the hardware and getting familiar with the Arduino IDE for controlling the ultrasonic sensor and servo motor.

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### Components Required:

1. **Arduino Uno** - Microcontroller board for controlling the system.
  2. **Ultrasonic Sensor (HC-SR04)** - For detecting objects and measuring distances.
  3. **Servo Motor** - For rotating the ultrasonic sensor to simulate radar scanning.
  4. **Breadboard** - For connecting components without soldering.
  5. **Jumper Wires** - For making electrical connections.
  6. **Arduino Cable (USB)** - For connecting the Arduino to a computer for programming and power.
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### Theory:

1. **Arduino Uno:**
  - A microcontroller board based on the ATmega328P.
  - It has 14 digital I/O pins, 6 analog inputs, and can be programmed using the Arduino IDE.
  - Used to control the ultrasonic sensor and servo motor.
2. **Ultrasonic Sensor (HC-SR04):**
  - Measures distance using sound waves.
  - Emits an ultrasonic pulse and listens for the echo to calculate the distance based on the time delay.

- Range: 2 cm to 400 cm.
  - 3. **Servo Motor:**
    - A motor that can rotate to a specific angle (0° to 180°).
    - Controlled using PWM (Pulse Width Modulation) signals from the Arduino.
  - 4. **Breadboard:**
    - A prototyping tool for building temporary circuits without soldering.
    - Allows easy connection of components using jumper wires.
  - 5. **Jumper Wires:**
    - Used to connect components on the breadboard and to the Arduino.
  - 6. **Arduino IDE:**
    - Software used to write, compile, and upload code to the Arduino board.
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## **Setup and Connections:**

1. **Connect the Ultrasonic Sensor to Arduino:**
    - VCC → 5V on Arduino
    - GND → GND on Arduino
    - Trig → Digital Pin 9 on Arduino
    - Echo → Digital Pin 10 on Arduino
  2. **Connect the Servo Motor to Arduino:**
    - VCC (Red Wire) → 5V on Arduino
    - GND (Black/Brown Wire) → GND on Arduino
    - Signal (Yellow/Orange Wire) → Digital Pin 6 on Arduino
  3. **Power the Arduino:**
    - Connect the Arduino to the computer using the USB cable.
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## Arduino Code:

java

```
#include <Servo.h>
```

```
// Defines Trig and Echo pins of the Ultrasonic Sensor
```

```
const int trigPin = 5;
```

```
const int echoPin = 6;
```

```
long duration;
```

```
int distance;
```

```
Servo myServo; // Creates a servo object
```

```
void setup() {
```

```
    pinMode(trigPin, OUTPUT);
```

```
    pinMode(echoPin, INPUT);
```

```
    Serial.begin(9600);
```

```
    myServo.attach(7); // Servo motor connected to pin 7
```

```
}
```

```
void loop() {
```

```
    // Rotates the servo motor from 15 to 165 degrees (slowly)
```

```
    for (int i = 15; i <= 165; i += 2) { // Moves in steps of 2 instead of 1
```

```
        myServo.write(i);
```

```
        delay(50); // Increased delay for slower movement
```

```

distance = calculateDistance();

Serial.print("Angle: ");

Serial.print(i);

Serial.print("°, Distance: ");

Serial.print(distance);

Serial.println(" cm");

}

// Rotates back from 165 to 15 degrees (slowly)

for (int i = 165; i > 15; i -= 2) { // Moves in steps of 2

    myServo.write(i);

    delay(50); // Increased delay for smoother movement

    distance = calculateDistance();

    Serial.print("Angle: ");

    Serial.print(i);

    Serial.print("°, Distance: ");

    Serial.print(distance);

    Serial.println(" cm");

}

// Function to calculate distance using Ultrasonic Sensor

```

```
int calculateDistance() {  
  
    digitalWrite(trigPin, LOW);  
  
    delayMicroseconds(2);  
  
    digitalWrite(trigPin, HIGH);  
  
    delayMicroseconds(10);  
  
    digitalWrite(trigPin, LOW);  
  
  
    duration = pulseIn(echoPin, HIGH);  
  
    distance = duration * 0.034 / 2;  
  
    return distance;  
  
}
```

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### **Procedure:**

1. Assemble the circuit as per the connections.
  2. Open the Arduino IDE and upload the provided code to the Arduino Uno.
  3. Open the Serial Monitor in Processing software (Ctrl+Shift+M) to observe the distance measurements at different angles.
  4. Verify that the servo motor rotates and the ultrasonic sensor detects objects within its range.
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### **Observations:**

- The servo motor should rotate from 0° to 180° and back, while the ultrasonic sensor measures distances.
  - The Serial Monitor will display the angle and corresponding distance of detected objects.
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### **Conclusion:**

This experiment introduces the basic components of a radar detection system and demonstrates how to control them using Arduino. The ultrasonic sensor measures distances, and the servo motor simulates a scanning radar. This setup forms the foundation for more advanced radar detection systems.

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### **Precautions:**

1. Ensure all connections are secure to avoid short circuits.
  2. Do not exceed the voltage limits of the components.
  3. Handle the servo motor gently to avoid damaging its gears.
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